



N-Type Bifacial Solar Panel

USER MANUAL V1.0.0



Model : CN120W/CN180W/CN200W/CN275W

IEC 61215, IEC 61730

ISO 9001: 2023 Quality management systems

ISO 14001: 2023 Environmental management systems

ISO 45001: 2023 Occupational health and safety management systems

Thank you for choosing our high-performance solar charging solution. We are dedicated to providing you with reliable off-grid power, and your safety is our top priority. To ensure optimal performance and protect your battery system, please take a moment to read this manual thoroughly before installation and operation. By using this product, you agree to follow the safety guidelines and operating instructions outlined in this guide to ensure a smooth and safe user experience.

To ensure long-term product reliability and maintain your warranty coverage, please handle and install the equipment with care. Please note that the manufacturer cannot accept responsibility for product failures, property damage, or injuries resulting from improper use, neglect, or unauthorized modifications. This includes, but is not limited to: intentional or accidental short-circuiting, exposure to extreme thermal environments, unauthorized disassembly, physical damage from heavy impacts or crushing, or using the panel with incompatible or low-quality charge controllers.

***COMPATIBILITY NOTE:** Please verify in advance that the voltage, current, and power parameters of this solar panel match the input specifications of your connected system. Failure to ensure compatibility may result in permanent damage to your devices or the solar panel.

Product Use Precautions

- 1. No Modifications:** Never attempt to disassemble, modify, or insert foreign objects into the unit, as this may cause unpredictable hazards and void the warranty.
- 2. Fire Safety:** Keep the solar panel away from flammable or combustible materials during charging.
- 3. Handling with Care:** Handle the unit with care. Avoid dropping, placing heavy loads on, or subjecting the panel to strong physical impacts.
- 4. Anti-Collision & Rigid Care:** Do not attempt to bend or flex the solar panel. Avoid collisions with hard surfaces or contact with sharp objects that could scratch or damage the protective layer.
- 5. No Stepping:** Do not step, stand, sit, or jump on the solar panel surface. Heavy pressure or sharp impacts can cause invisible micro-cracks in the solar cells, severely reducing power output or creating safety hazards.
- 6. Cable Protection:** Do not pull or yank the power cables forcefully. Always grasp the connector plug when disconnecting to prevent cable breakage, internal damage, or loose connections.

COMPATIBLE MODELS

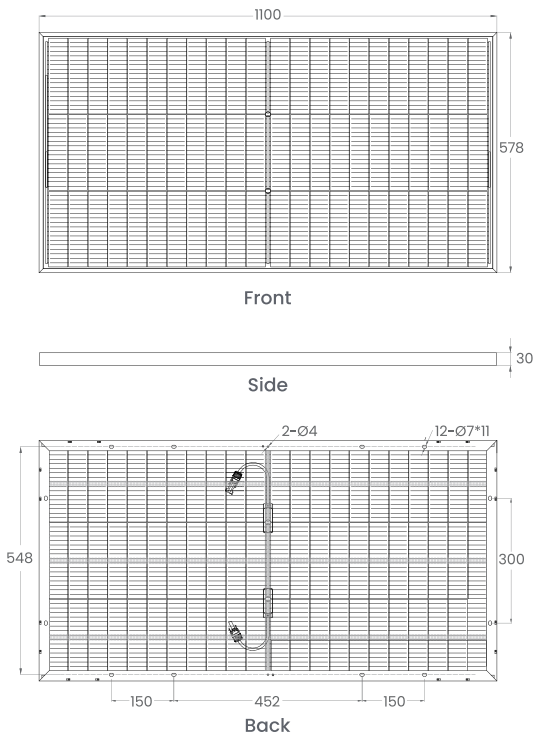
120W N-type Bifacial Solar Panel (CN120W)
180W N-type Bifacial Solar Panel (CN180W)
200W N-type Bifacial Solar Panel (CN200W)
275W N-type Bifacial Solar Panel (CN275W)

Product Highlights

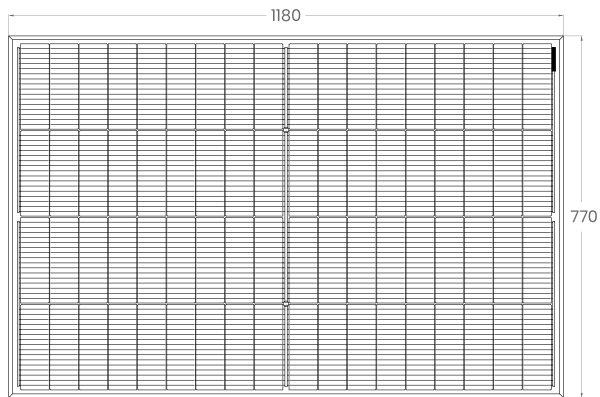
 Strengthened Mechanical Load Professionally reinforced to handle high mechanical loads for ultimate safety assurance.	 Safety Assurance Under Extreme Heat Features excellent fire resistance and structural stability under extreme heat.
 Innovative Bifacial Design Harvests solar energy from both sides to keep your system fully charged naturally.	 Excellent Anti-PID Performance Guaranteed through an optimized production process and strict material control.
 SMBB (Super Multi-Busbar) Technology Minimizes power loss to create tougher and more efficient solar modules.	 Innovative Dual-Matrix Parallel Design Ensures that shading on one half will not impact the power output of the other.

Product Dimensions

Model: CN120W
1100 × 578 × 30 (±5mm)



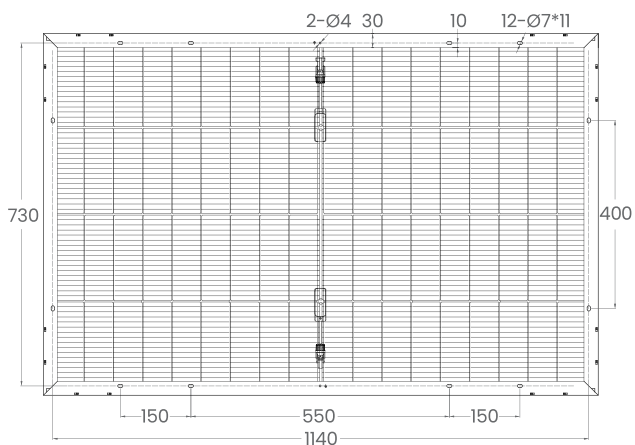
Model: CN180W
 1180 × 770 × 35 (±5mm)



Front



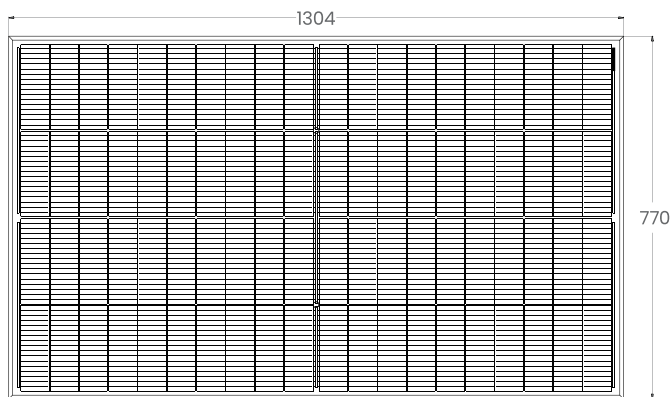
Side



Back

CN200W

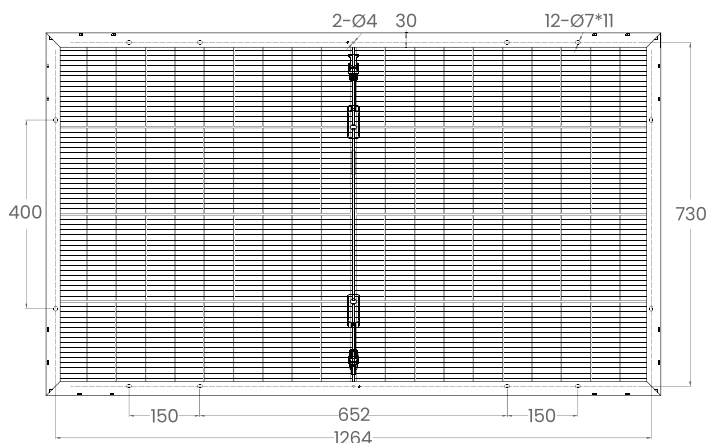
1304 × 770 × 35 (±5mm)



Front



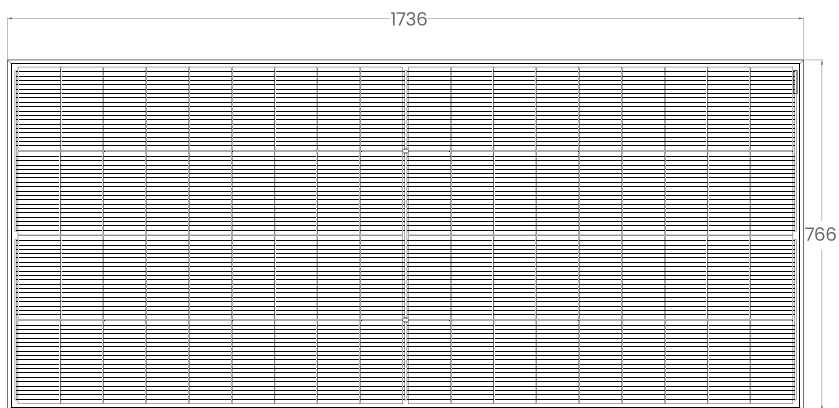
Side



Back

Model: CN275W

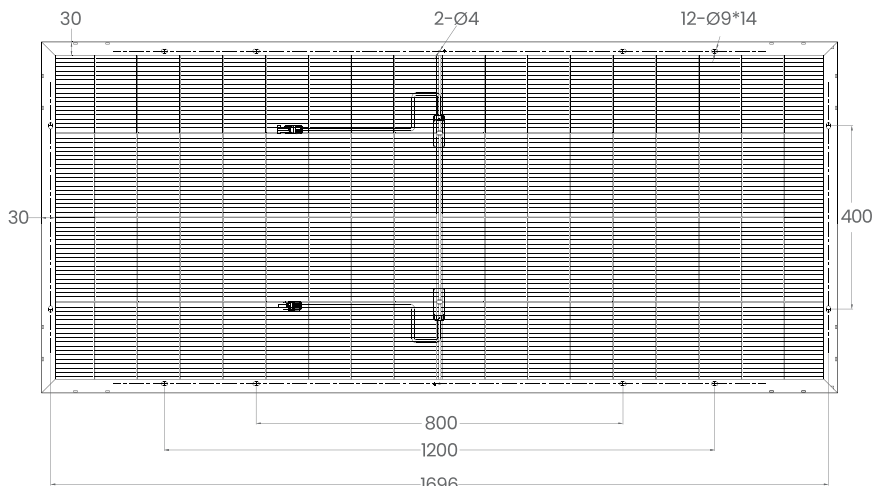
1736 × 766 × 35 (±5mm)



Front



Side



Back

Specifications

Model	CN120W	CN180W	CN200W	CN275W
Maximum Power at STC (P _{max})	120W	180W	200W	275W
Maximum Power Voltage at STC (V _{mp})	20.48V	21.93V	23.74V	21.29V
Maximum Power Current at STC (I _{mp})	5.86A	8.21A	8.43A	12.92A
Open-circuit Voltage at STC (V _{oc})	24.06V	25.56V	27.31V	25.26V
Short-circuit Current at STC (I _{sc})	6.21A	8.62A	8.91A	13.69A
Cell Efficiency	25.6%			
Power Tolerance	0-5%			
Maximum System Voltage	1000V DC			
Maximum Series Fuse Rating	25A			
First Year Degradation Rate	1%			
Linear Degradation Rate	0.4% / year			

*STC(Standard Test Condition): Irradiance 1000W/m², Cell Temperature 25°C, Air Mass 1.5.

Mechanical Characteristics

Model	CN120W	CN180W	CN200W	CN275W
Dimensions	1100 × 578 × 30 (±5mm)	1180 × 770 × 35 (±5 mm)	1304 × 770 × 35 (±5 mm)	1736 × 766 × 35 (±5 mm)
	43.31 × 22.76 × 1.18 (±0.2 in)	46.46 × 30.31 × 1.38 (±0.2 in)	51.34 × 30.31 × 1.38 (±0.2 in)	68.35 × 30.16 × 1.38 (±0.2 in)
Weight	6.85 (±0.2 kg)	9.80 (±0.2 kg)	10.80 (±0.2 kg)	13.51 (±0.2 kg)
	15.09 (±0.5 lb)	21.61 (±0.5 lb)	23.81 (±0.5 lb)	29.78 (±0.5 lb)
Glass	3.2mm Tempered Glass			
Frame	Anodized Aluminium Alloy			
IP Rating	IP68			
Max. Wind Load/ Snow Load	2400Pa / 5400Pa			
Connectors	Solar Connectors			
Cell Type	N-Type Monocrystalline Silicon			

Temperature Coefficients

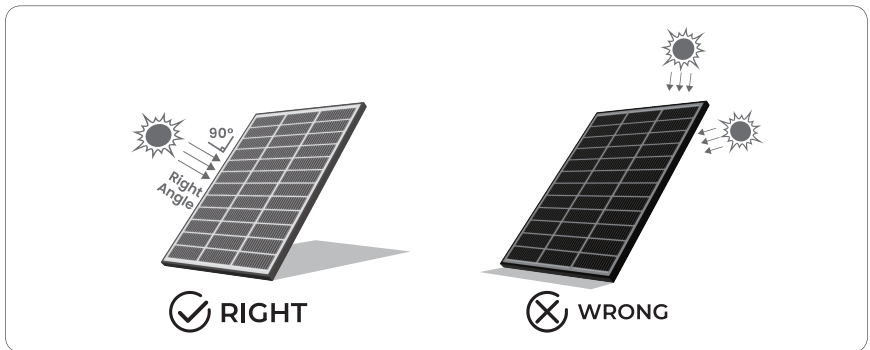
Model	CN120W	CN180W	CN200W	CN275W
Temperature coefficients of P _{max}	-0.30%/K			
Temperature coefficients of V _{oc}	-0.25%/K			
Temperature coefficients of I _{sc}	0.045%/K			
Operating Temperature(°C)	-40°C ~ 85°C			

I. Quick Charging & Operation Guide

- 1. Daytime Charging Only:** Solar panels only produce power when exposed to sunlight. However, you can safely leave the panel connected overnight without unplugging it, as it features built-in reverse-current protection. For best results, rely on peak daylight hours.
- 2. Verify Device Limits:** Verify Device Limits: Always check the PV input voltage and wattage limits of your connected system before connecting. If your connected system has a fixed input cap, it will only charge up to that maximum limit, even if the solar panel is capable of producing more.
- 3. Handle with Care:** Avoid physical impacts, heavy pressure, dropping, or stepping on the solar panel to prevent internal cell damage.

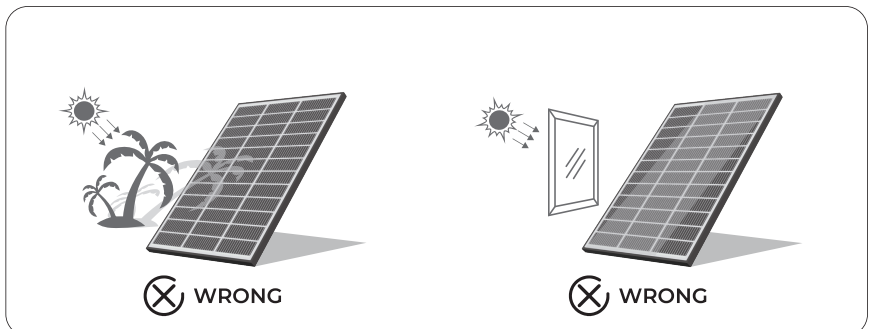
Optimize Sun Alignment

For maximum energy absorption, adjust the tilt angle of the solar panel so that its surface is **perpendicular** to the incoming sunlight.



Avoid Obstructions & Shading

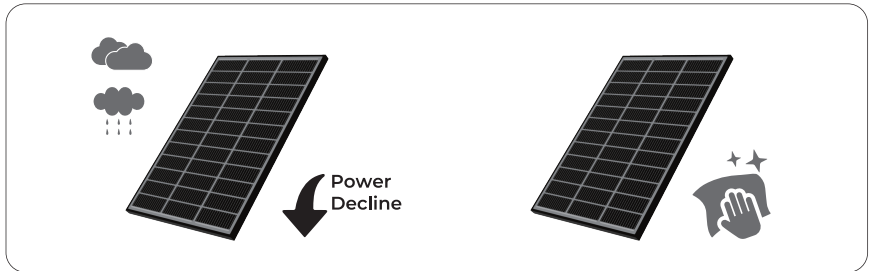
Always use the solar panel outdoors in an open area. Do not place it near trees or buildings. Avoid using the panel behind glass, windows, curtains, or screens, as filtered light will drastically reduce or eliminate power output.



II. Environmental Factors Affecting Output

Weather Conditions: Cloud cover, haze, rain, or snow will naturally reduce the solar panel's power generation.

Surface Cleanliness: Dust, dirt, or debris on the panel surface will block light absorption and lower efficiency. Please clean the surface regularly with a soft cloth.



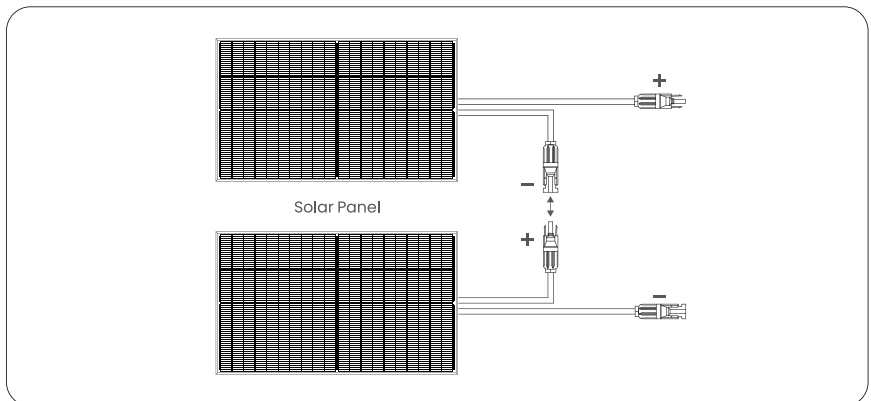
III. Multi-Panel Connection Guide (Series & Parallel)

Crucial Rules: For optimal performance and safety, always connect solar panels of the **same brand and identical specifications** to ensure balanced electrical performance.

Series Connection (Increases Voltage):

* Principle: Voltages add up while current remains the same. Total system current is restricted by the panel with the lowest output.

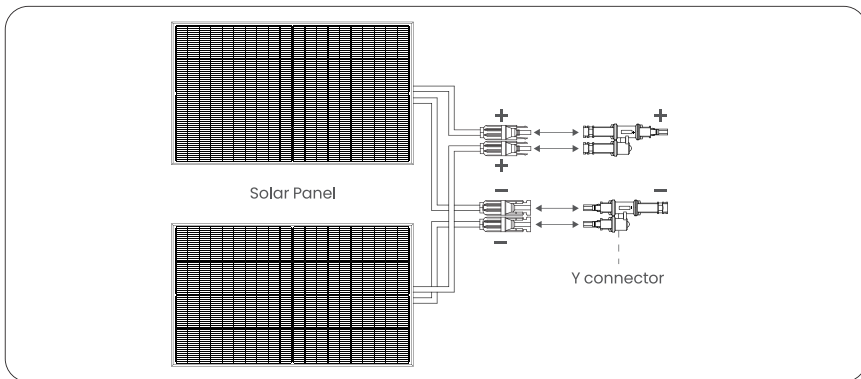
How to Connect: Plug the positive (+) solar connector of the first panel directly into the negative (-) solar connector of the second panel.



Parallel Connection (Increases Current):

* Principle: Currents add up while voltage remains the same. Total system voltage is restricted by the panel with the lowest voltage.

How to Connect: Use an Y-branch Connector (not included). Connect the positive (+) terminals of both panels to one Y-branch connector, and the negative (-) terminals of both panels to the other Y-branch connector.



※ Note: When combining panels in a series-parallel configuration, always double-check your solar charge controller's maximum voltage and current limits to prevent equipment damage.

Video Guide Available: For step-by-step visual instructions on how to connect your solar panels in series or parallel, please feel free to reach out to our customer service at: callsun-service@outlook.com. We're here to help!

Product Maintenance

Over time, dust, pollen, bird droppings, or debris may accumulate on the panel surface, blocking sunlight and reducing energy conversion efficiency.

- **Cleaning Method:** Periodically clean the solar panel by gently wiping the surface with a soft, damp cloth or sponge and fresh water.
- **WARNING:** Do not use abrasive pads, wire brushes, or corrosive chemical liquids, as they will permanently scratch the high-transmittance glass coating. Never use a high-pressure power washer, as it may damage the sealants or junction box.

Suitable Environments

For maximum power output, the solar panel matrix must be placed in an open outdoor area with unobstructed, direct sunlight.

- **Avoid Shading:** Do not operate the panel in areas shadowed by clouds, trees, power poles, or nearby buildings, as even partial shading will bottleneck the output of the entire array.
- **Weather & Temperature:** This solar panel is engineered to withstand diverse weather, but optimal charging occurs under clear, unobstructed skies. In winter, ensure the surface is kept free of heavy snow accumulation.

Warranty Policy

At Callsun, we stand behind the craftsmanship and long-term reliability of our products. This model features a **10-year limited warranty** covering any defects in materials and workmanship under standard operational conditions, beginning from your date of purchase. To help us serve you better, please retain your purchase receipt as valid proof.

What is kindly excluded from this warranty

To ensure fairness and transparency, please note that our warranty does not cover:

1. Normal wear and tear, or gentle cosmetic changes (such as minor surface scratches or slight color fading from regular outdoor sun exposure) that do not affect the solar panel's power generation.
2. Malfunctions resulting from unauthorized modifications, self-disassembly, field repairs, or accidental physical damage (such as dropping or heavy crushing).
3. Damages caused by force majeure or unpredictable natural factors, including lightning strikes, extreme hail, floods, severe storms, or accidents.
4. Connecting the panel to sub-standard, defective, or electrically incompatible controllers and devices.

How We Assist You (Warranty Process):

1. Reach Out: Simply contact our customer support team with your purchase proof and a brief description (along with photos or videos) of the issue you are experiencing.
2. Evaluation: Our dedicated technical team will carefully review your details within 24 business hours to understand the root cause.
3. Our Commitment: Once a manufacturing fault is identified, we will quickly provide you with a tailored, hassle-free solution (such as a replacement or repair guidance) to get your system back up and running.

Questions or Need a Hand?

Thank you for bringing Callsun into your life. For any product inquiries, setup assistance, or friendly technical support, please don't hesitate to contact our customer care team at: **callsun-service@outlook.com**

*Specifications are subject to change without notice to ensure continuous product improvement.

Shenzhen Guangchuxinneng Technology Co., Ltd.

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